

Calculating gain for RF

(1) Flip Angle

$$\alpha = B \times \tau \times \lambda$$

τ = pulse duration

λ = gyromagnetic ratio $\approx 42.58 \text{ MHz/T}$

α = can assume 90° or Ernst angle

Ernst angle

$$\alpha = \cos^{-1}\left(e^{-\frac{TR}{T_1}}\right)$$

TR = Time between RF pulses.

T_1 = longitudinal relaxation
Time of atom

(2) Solenoid Current

$$I = \frac{B L}{\mu_0 N}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

$$N = 20$$

$$L = 200 \text{ mm}$$

$\rightarrow$ plug (1) into (2)

$$I = \frac{\left(\frac{\alpha}{\tau \lambda}\right) L}{\mu_0 N}$$

(3) Voltage/gain

$$V_2 = I Z \rightarrow \text{assume } Z = 50 \Omega, (\text{perfect tuning})$$

$$V_2 = 50 \left(\frac{\left(\frac{\alpha}{\tau \lambda}\right) L}{\mu_0 N} \right)$$

$$\frac{V_2}{V_1} \geq 54 \text{ dB}$$

minimum gain
for Amp.